

Editor-in-Chief

IEEE Journal of Translational Engineering in Health and Medicine (JTEHM)
IEEE Engineering in Medicine and Biology Society

Dear Editor-in-Chief,

We submit for your consideration the manuscript “**Geodesic Diagnostic Trajectories (GDT): A Differential Geometry Framework for Adaptive Decision Support and Operational Optimization in EHR Systems**” by Hemanth Manchabale Papachappa and Aniruddha Ganguly, for review as a regular article in *IEEE Journal of Translational Engineering in Health and Medicine*.

Clinical problem and translational motivation. Chart review in electronic health record (EHR) systems is a multi-pivot, cognitively demanding task. A physician tracking a critically ill patient may cycle through culture results, renal dosing adjustments, and anticoagulation orders within a single session—each pivot rendering the prior retrieval context a source of interference rather than assistance. Standard session-based retrievers accumulate this stale context without detection, degrading result relevance at the moment clinical decision speed matters most. GDT is our direct engineering response to this failure mode, designed to move from academic benchmark toward real-time clinical deployment.

Technical contribution. GDT encodes the clinician’s evolving diagnostic intent as a trajectory on the Riemannian manifold $\mathcal{S}_d^+$ of symmetric positive definite (SPD) matrices. A geodesic shift ratio κ_t quantifies abrupt topic pivots relative to recent trajectory pace; a soft gate driven by κ_t suppresses stale context with suppression strength bounded by a formally proved proposition (Proposition 1). A JEPA-style latent predictor anticipates the next query and populates a prefetch cache; prefetch volume is scaled by trajectory confidence so that uncertain post-pivot forecasts do not contaminate the cache. An M/M/c queueing model connects the resulting latency gains to ward-level operational throughput estimates.

Clinical benchmark and key findings. We evaluated GDT against CMA, BM25, and TF-IDF in a randomized four-arm within-subject crossover study spanning 124 sessions on 31 multi-pivot clinical vignettes from MIMIC-III/IV, eICU, and HiRID. GDT reduced mean retrieval latency by 26.9% relative to TF-IDF control ($p < 0.001$, $d = -4.12$). Proposition 1 held across all 1,074 gate-triggered transitions without exception. The M/M/c projection estimates a 4.1-minute reduction in per-reviewer patient processing time during peak-acuity shifts. Equally important is the finding that all four conditions achieved 0% task completion—an honest diagnostic result that isolates the vocabulary mismatch inherent in sparse lexical retrieval and identifies dense retrieval integration as the necessary next step.

Fit for JTEHM. This manuscript aligns directly with JTEHM’s mission of translational engineering for health and medicine. The contributions span three translational dimensions: (1) a theoretical framework with a proved interference bound—the kind of formal guarantee that regulatory bodies and clinical governance committees increasingly expect from deployed CDS; (2) a simulation benchmark that makes the system’s limitations explicit rather than obscuring them; and (3) an operational analytics model that translates sub-second engineering improvements into workflow metrics meaningful to hospital administrators. The paper also discusses concrete deploy-

ment prerequisites—SMART on FHIR integration, NASA-TLX human-subject validation, fairness auditing—situating it firmly in the translational rather than purely algorithmic literature.

Confirmations.

- This work has not been published and is not under consideration elsewhere.
- All authors have approved the manuscript and agree with the submission.
- All datasets used (MIMIC-III/IV, eICU, HiRID) are publicly available and de-identified; no new protected health information was collected.
- The evaluation pipeline will be shared on GitHub upon acceptance.
- There are no conflicts of interest to disclose (see accompanying statement).

We believe GDT’s combination of formal theoretical foundations, transparent reporting of null retrieval-quality findings, and operational throughput modeling makes it a timely and relevant contribution to JTEHM’s readership. We look forward to the editors’ and reviewers’ feedback.

Sincerely,

Hemanth Manchabale Papachappa
Aniruddha Ganguly